

DEEP LEARNING RESEARCH ANALYSIS

Syllable-Level Stress Classification in L1 Sindhi & L2 English

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PART 1: THE CONCEPTUAL LAYER

Why sequential modeling is essential for phonetic prosody.

MODELING TEMPORAL CONTEXT

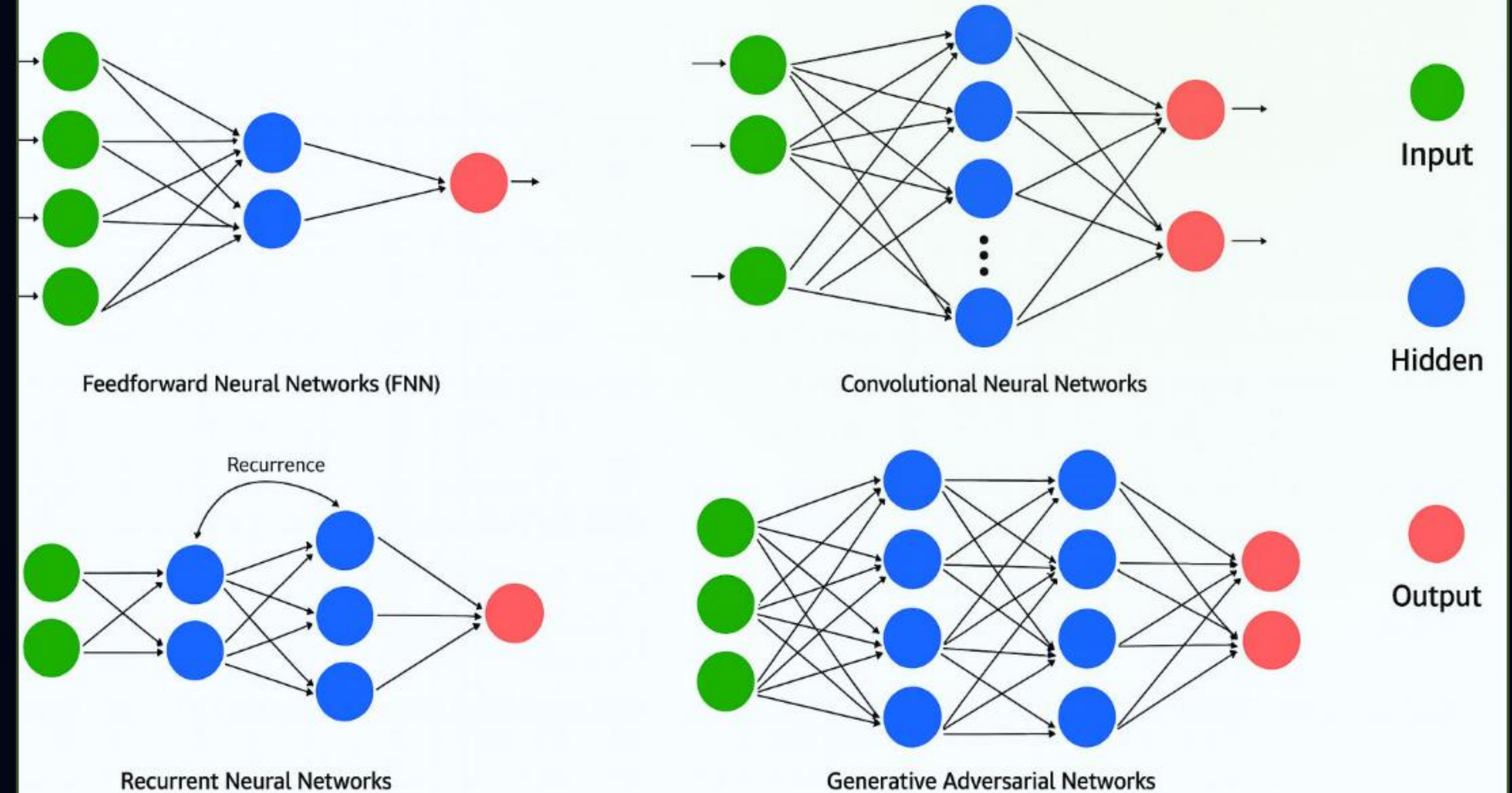
Speech is Sequential Data: A syllable's stress is defined by its relationship to surrounding neighbors.

Forward Context: Captures rhythmic flow leading into the syllable.

Backward Context: Captures the "rebound" or decay energy after the syllable.

BiLSTM Utility: Concatenates both hidden states to form a holistic phonetic window.

Different Types of Neural Network Architectures



PART 2: THE MATHEMATICAL CORE

Optimization, Partial Derivatives, and Loss Functions.

BINARY CROSS-ENTROPY LOSS

For Stress (1) vs. Unstressed (0) classification, the model minimizes the Negative Log-Likelihood:

$$L = -\frac{1}{N} \sum_{i=1}^N [y_i \log (\hat{y}_i) + (1 - y_i) \log (1 - \hat{y}_i)]$$

Where $\hat{y}_i$ is the probability output from the final Sigmoid layer.

TRAINING CONVERGENCE



The model reaches convergence rapidly due to the rich initialization from Wav2Vec 2.0 embeddings.

PART 3: FEATURE ENGINEERING

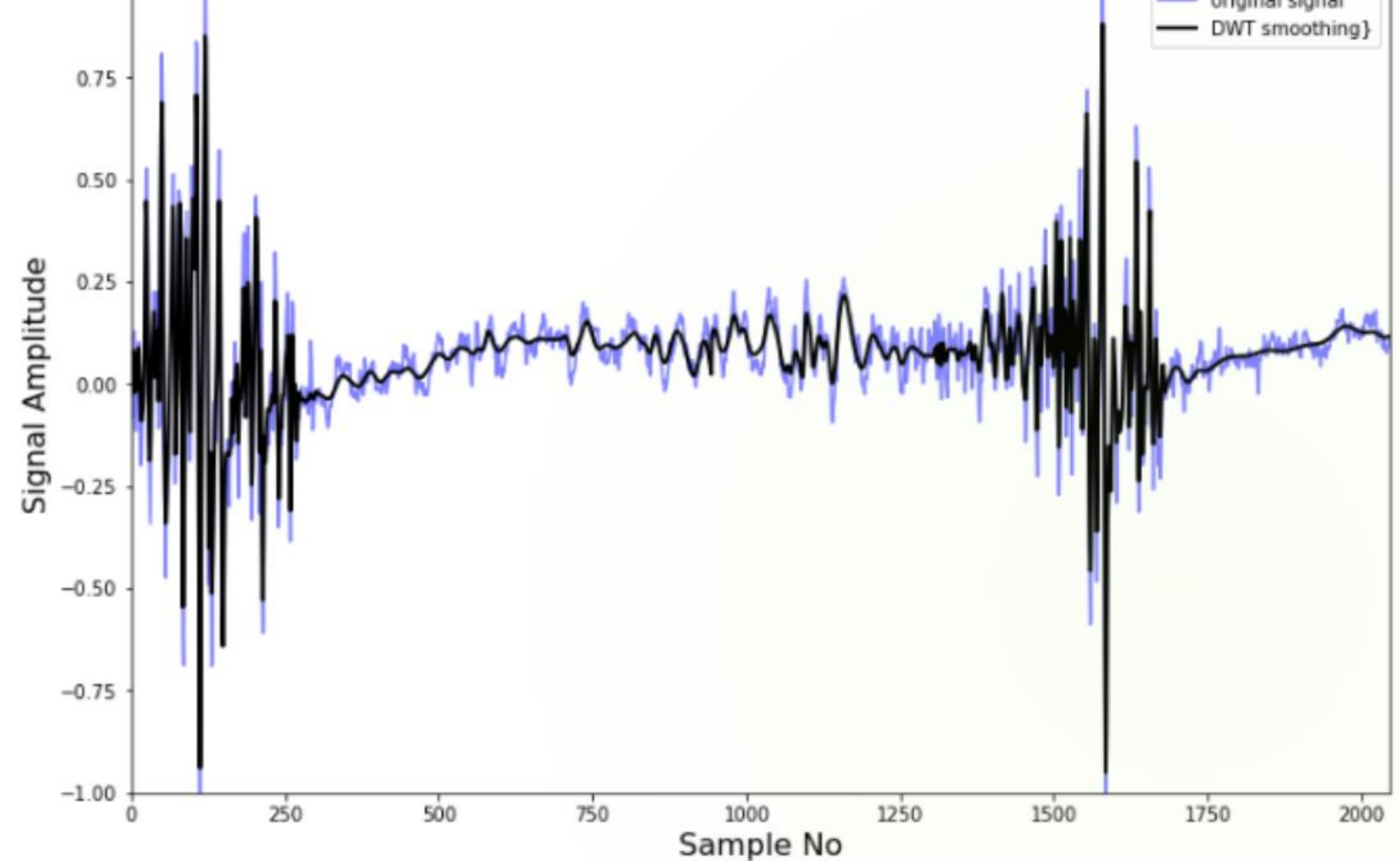
From Raw Audio to Prominence Spectrograms.

SIGNAL DECOMPOSITION

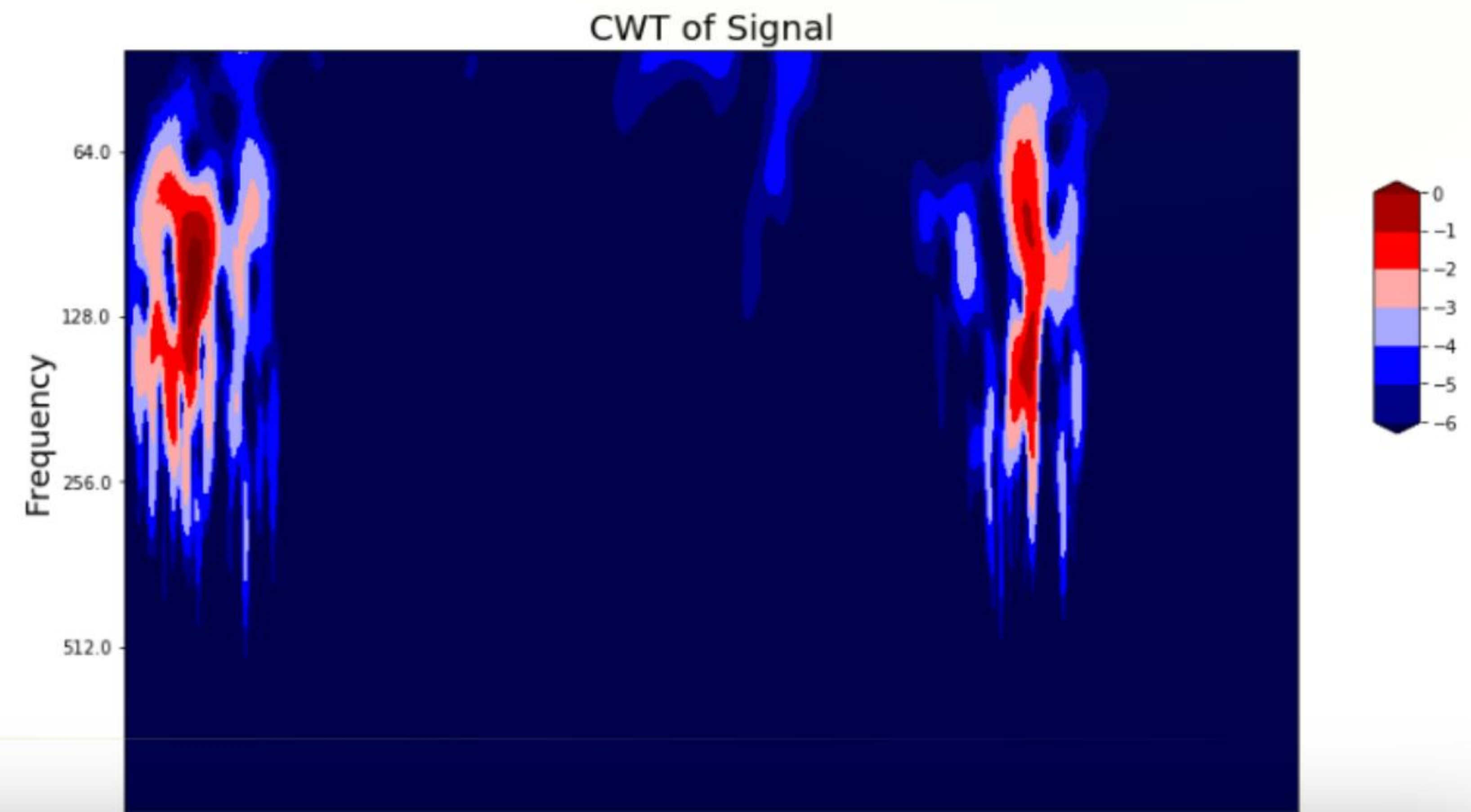
Unlike FFT, the Continuous Wavelet Transform uses a "Mother Wavelet" scaled by a and shifted by b :

$$T(a, b) = \frac{1}{\sqrt{a}} \int x(t) \psi^* \left(\frac{t-b}{a} \right) dt$$

This creates a Spatial Feature Map that BiLSTM processes similarly to CNN spatial pixel data.



coeff shape is: (48, 2048)
frequency shape is: (48,)



MODEL PERFORMANCE RESULTS

79.2%

Traditional
MFCC + ANN

84.5%

Basic LSTM
(Uni-dir)

89.1%

Fused
CWT-Wav2Vec

92.1%

**BiLSTM-Hybrid
(Proposed)**

Softmax activations for final syllable probabilities ensure high precision in stress detection.

BENCHMARK SUMMARY

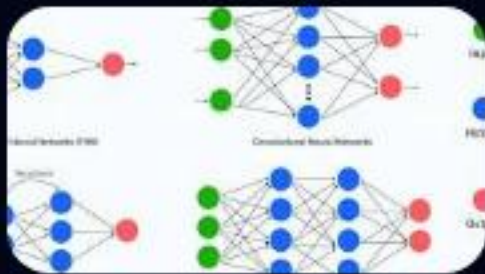
Architecture	Feature Set	Accuracy
ANN	MFCC / Spectral	79.2%
LSTM	Time-Domain	84.5%
Proposed Hybrid	Wavelet + Wav2Vec	92.1%

QUESTIONS?

Applying ANN & Deep Learning to Regional Prosody Analysis

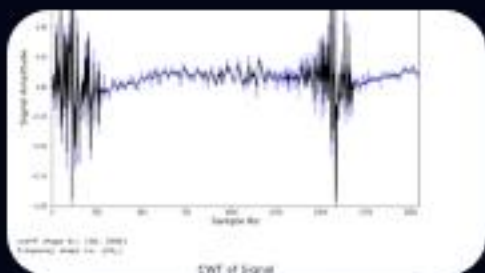
SMIU, KARACHI

IMAGE SOURCES



https://ik.imagekit.io/upgrad1/abroad-images/imageCompo/images/ChatGPT_Image_Nov_20_2025_03_51_24_PMEIN42V.png

Source: www.upgrad.com



<https://i.sstatic.net/qUYpj.png>

Source: dsp.stackexchange.com



<https://www.smiu.edu.pk/themes/smiu/images/l2.jpg>

Source: www.smiu.edu.pk
